

Separation of all oligomers in polyethylene glycols and their monomethyl ethers by one-dimensional liquid chromatography

Bernd Trathnigg · Hasnat Ahmed

Received: 26 April 2010 / Revised: 18 June 2010 / Accepted: 20 June 2010 / Published online: 8 July 2010
© Springer-Verlag 2010

Abstract Different polymer homologous series having the same repeat unit, but different end groups, can be separated by one-dimensional LAC according to the number of repeat units and functionality, if a favorable combination of the interaction parameters of the repeat unit and the end groups can be found. As an example, polyethylene glycol (PEG) can be determined in PEG monomethyl ethers. The molar mass distribution of the minor component in such samples can be determined even at concentrations of a few percent.

Keywords Polyethylene glycol · Monomethyl ether · Functionality · Liquid adsorption chromatography · Interaction parameter

Introduction

The presence of chains having a different functionality and their determination is a pertinent problem in polymer chemistry. For some applications, small amounts of such impurities can be negligible, but even traces of chains with different functionality may have a dramatic effect on product quality. Polymers and oligomers with different functionality may be formed as side products by chain transfer reactions in the polymerization; they may, however, also originate from incomplete conversion in the modification of end groups.

In the ring-opening polymerization of cyclic ethers and lactones, cycles may be formed besides the desired linear

chains [1]. In the synthesis of monofunctional polyethers by ring-opening polymerization of epoxides with an alcoholate as initiator diols may be formed by chain transfer to traces of water. On the other hand, chain transfer to the monomer may occur in the polymerization of propylene oxide, which results in the formation of chains with allylic end groups besides the desired diols [2–5]. If the diols are used as chain extenders (e.g., in the production of polyurethanes), monofunctional molecules will have an adverse effect on the properties of the final product.

The same reactions can also occur in the synthesis of block copolymers: traces of water in the polyethylene glycol (PEG) used as initiator will result in the formation of homopolymers by a chain transfer reaction.

If a chain with terminal hydroxy groups is esterified, one may expect unreacted starting material besides mono- and diesters. Depending on the application, this will have different consequences: Monoester of fatty acids with PEG can be used as emulsifier [6, 7]. Obviously, PEG has no emulsifying properties; hence, it is useless, but the diesters are highly hydrophobic and would even require an emulsifier to prevent precipitation. A much more dramatic effect must be expected in macromonomers, from which graft copolymers can be synthesized by copolymerization with other monomers: The macromonomers are typically synthesized by reaction of PEG–monomethyl ethers (MME) with methacryloyl chloride [8]. If the macromonomer contains diesters, these will act as crosslinker.

A similar effect must be expected in the PEGylation of proteins and drugs etc. [9–15]. For this purpose, monomethyl ethers of PEG are typically used, which can be synthesized by polymerization of ethylene oxide with methanol or the monomethyl ether of triethylene glycol [8] as initiator and sodium hydroxide or a similar catalyst (such as potassium naphthalene [16]). The presence of PEG

Published in the special issue on *Separation Science of Macromolecules* with Guest Editor André Striegel.

B. Trathnigg (✉) · H. Ahmed
Institute of Chemistry, Karl-Franzens-University,
Central Polymer Laboratory/Molecular Characteristics (CePol/MC),
8010 Graz, Austria
e-mail: bernd.trathnigg@uni-graz.at

having two active sites results in cross linking of two proteins or even the formation of aggregates [17].

As the molar concentration of chains is much more important than the weight concentration, a reliable characterization of functional polymers and oligomers with respect to functionality and molar mass is crucial for many applications.

The question is, however, the suitability of the analytical methods. In literature, this has often been done by different chromatographic techniques, nuclear magnetic resonance (NMR), and MALDI-TOF-MS [8, 16].

All these techniques have some limitations. For example, gas chromatography can be used only for separation of low molecular weight oligomers, and NMR yields only average values; moreover, it is not sensitive enough for the determination of low contents [17]. MALDI-TOF-MS can perfectly resolve all oligomers, but is problematic with respect to quantitation, especially in the low molecular range (close to the molar mass of the matrix).

In principle, a separation of polymers according to functionality can be achieved by liquid chromatography at critical conditions (LCCC) [8, 18–21]. At a special mobile phase composition and temperature, the retention of a polymer chain is independent of molar mass. At this so-called critical point of adsorption (CAP), a separation according to functionality becomes possible. This technique works very well with strongly adsorbing end groups, as is the case with fatty alcohol ethoxylate or alkyl phenol ethoxylates, but it is difficult if the end groups and the repeat units are not sufficiently different.

Such a situation occurs with PEG-MME: At the CAP for PEG on a reversed-phase column, the methoxy end group of PEG-MME is adsorbed very slightly. Nevertheless, such a separation has been described recently [22]. We have thus decided to concentrate on this system as a commercially important example.

Besides the total amount of the by-product PEG (which can be obtained from LCCC), its molar mass distribution (MMD) is important as the amount of diols depends on the total amount of the PEG and its number average of molar mass.

The molar mass of the PEG is also highly important for another reason: In the separation described by Barman, an evaporative light scattering detector (ELSD) has to be used, as PEG elutes with the solvent peak. With this detector, one has to take into account not only its nonlinear response but also the molar mass dependence of response factors [23]. Barman obtained this information by LC-MS, which is, however, not available in most labs. Moreover, quantification of small amounts is somewhat problematic in LC-MS. The better approach would be a separation of all oligomers of both series (glycols and monomethyl ethers) by chromatography.

In principle, this is possible by two-dimensional liquid chromatography (2D-LC). As we have shown previously, a full separation of all oligomers of both homologous series can be achieved [24]. In this paper, we have described a new approach, which employs LCCC in the first dimension with heart-cutting of the PEG and PEG-MME fractions, which are subsequently analyzed by gradient LAC in the second dimension. The fractions are focused by adding water to the eluate from the first dimension (between the column and the switching valve) in order to make the solvent containing the fractions a weaker eluent than the mobile phase of the second dimension. This procedure works very well even at very low PEG content (less than 1%), but requires rather complex equipment.

Another approach could be LAC under conditions, which allow a separation of the individual polymer homologous series without overlapping peaks. In this study, we wanted to find conditions, under which such a separation is possible, and—if so—which amounts of PEG could be identified and quantitatively determined. Moreover, this approach should be compared with the others mentioned above.

Theoretical background

Retention of polymer homologous series in LAC can be described by Martin's rule, which is commonly used in the literature [25–29].

$$\ln k = A + Bn \quad (1)$$

In Eq. 1, k is the retention factor: $k = \frac{V_e - V_0}{V_0}$, V_e is the elution volume of an oligomer with n repeat units, and V_0 is the hold-up volume. Many chromatographers consider the elution volume of the solvent peak as the hold-up volume, which leads to considerable deviations from linearity at low n . The hold-up volume is not necessarily identical with the void volume, which is the total amount of mobile phase in the column, i.e., the sum of the interstitial volume V_i and the pore volume V_p : $V_0 = V_i + V_p$.

As has been shown previously [30–32], Martin's rule can be explained using a theory of polymer chromatography [21, 33–36]. At sufficiently strong interaction (which is typical for LAC), retention of a non-functional polymer with n repeat units can be described by

$$\ln k^* = \ln \left(\frac{4}{cD} \frac{V_p}{V_0^*} \right) + n \frac{c^2 a^2}{6} \quad (2)$$

wherein D is the pore diameter of the stationary phase (measured in nm), a is the length of the repeat unit (in nm), and c is the interaction parameter, which describes the interaction of the repeat unit with the surface of the

Table 1 Parameters obtained for PEG on Prodigy ODS3 in acetone–water with 12.04 wt.% acetone

$a=0.5$ nm
$c=2.1$ nm ⁻¹
$S_p=220$ m ²
$V_0^*=2.1$ mL

stationary phase. It is measured in inverse length (nm⁻¹) and does not depend on D and n .

In Eq. 2, the retention factor is given by $k^*=(V_e-V_0^*)/V_0^*$, wherein V_0^* is the accessible volume:

$$V_0^* = V_i + V_p \left(1 - \frac{4R}{D\sqrt{\pi}} \right) - \frac{2V_p}{cD} \quad (3)$$

Evidently, the accessible volume V_0^* is always smaller than the void volume $V_0=V_i+V_p$.

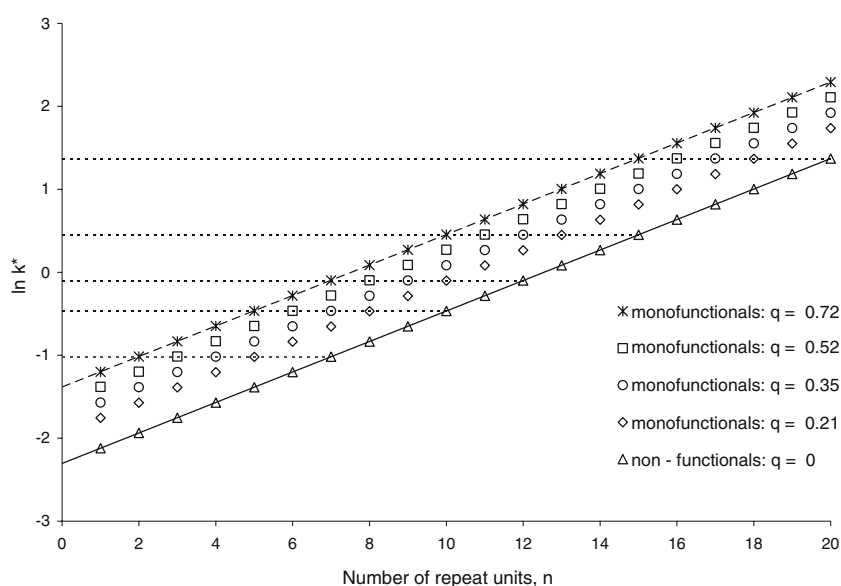
The radius of gyration R can be expressed by the length a and the number n of repeat units:

$$R = a\sqrt{\frac{n}{6}} \quad (4)$$

Obviously, Eq. 2 corresponds to Martin's rule (Eq. 1). The main difference is that perfectly straight lines are obtained, if the retention factor k^* is calculated using the accessible volume instead of the void volume.

As has been shown previously [31], the accessible volume in LAC can be easily obtained from a plot of the elution volumes V_n of oligomers with n repeat units versus the differences $\Delta V_n = V_n - V_{n-1}$ in the elution volumes of consecutive oligomers.

$$V_n = V_0^* + \gamma \Delta V_n \quad (5)$$

Fig. 1 Simulated retention using the parameters from Table 1: complete overlap of both series

The intercept in such a plot represents the accessible volume, and the interaction parameter c can be calculated from the slope $\gamma = \frac{e^B}{e^B - 1}$ (wherein $B = \frac{c^2 a^2}{6}$ is the slope in Martin's rule).

$$c = \frac{1}{a} \sqrt{6 \ln \frac{\gamma}{\gamma - 1}} \quad (6)$$

It must be mentioned that the determination of the accessible volume and the interaction parameter does not require the knowledge of n for the individual peaks (provided that they belong to the same series).

For monofunctional chains, an additional parameter has to be taken into account [34, 37]: The influence of specific groups can be expressed by the parameter $q = \delta(e^{\Delta\chi/RT} - 1)$.

The specific end group parameter q , which measures the difference of the free energy of adsorption $\Delta\chi$ (in RT units) between end group and repeat unit (δ is the interaction distance between the specific group and the sorbent), was discussed in previous papers [21, 35].

Retention of monofunctionals can be described by

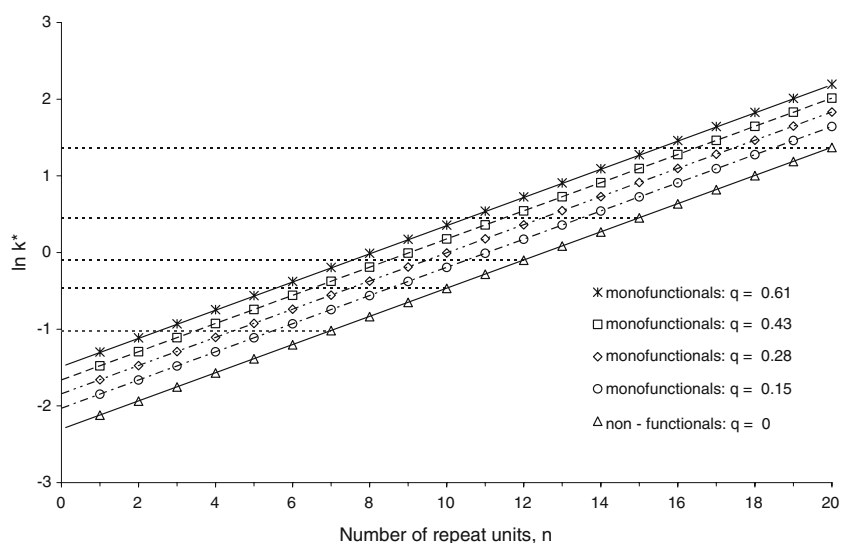
$$\ln k_m^* = \ln \left(\frac{4}{cD} \frac{V_p}{V_{0,m}^*} (1 + qc) \right) + n \frac{c^2 a^2}{6} \quad (7)$$

wherein $V_{0,m}^*$ is the accessible volume for monofunctionals, which is somewhat smaller than the accessible volume for non-functionals:

$$V_{0,m}^* = V_0^* - \frac{V_p}{D} \frac{2}{\sqrt{\pi}} \frac{q}{Rc}, \quad (8)$$

A simple procedure for the determination of q has been described previously [38]. From the elution volumes of the peaks of monofunctional and non-functional oligomers (V_m and V_n , respectively) with the same number of repeat units

Fig. 2 Simulated retention using the parameters from Table 1: perfect separation of both series



and the corresponding accessible volumes, one may calculate q (c is obtained from the same chromatograms using Eq. 5).

$$\frac{(V_m - V_{0,m}^*)}{(V_n - V_0^*)} = 1 + qc \quad (9)$$

In the case of weakly adsorbing end groups (such as the methoxy group in PEG-MME), the difference between $V_{0,m}^*$ and $V_{0,n}^*$ is typically rather small, but not negligible.

One may write Eq. 7 also in the form

$$\ln k_m^* = \ln \left(\frac{4V_p}{cDV_{0,m}^*} \right) + \ln(1 + qc) + n \frac{c^2 a^2}{6} \quad (10)$$

The difference in $\ln k^*$ of oligomers with the same n is given by

$$\ln k_m^* - \ln k^* = \ln \left(\frac{V_{0,n}^*}{V_{0,m}^*} \right) + \ln(1 + qc) + \Delta n \frac{c^2 a^2}{6} \quad (11)$$

On the other hand, the monofunctional oligomer with n repeat units will have the same retention factor as the non-functional oligomer with $n + \Delta n$ repeat units.

$$\Delta n = \frac{\ln \left(\frac{V_{0,n}^*}{V_{0,m}^*} (1 + qc) \right)}{B} \quad (12)$$

Fig. 3 LAC of PEG 600 and PEG-MME 350 on the Prodigy ODS3 column in 12.08 wt.% acetone at 25 °C. Detection: ELSD

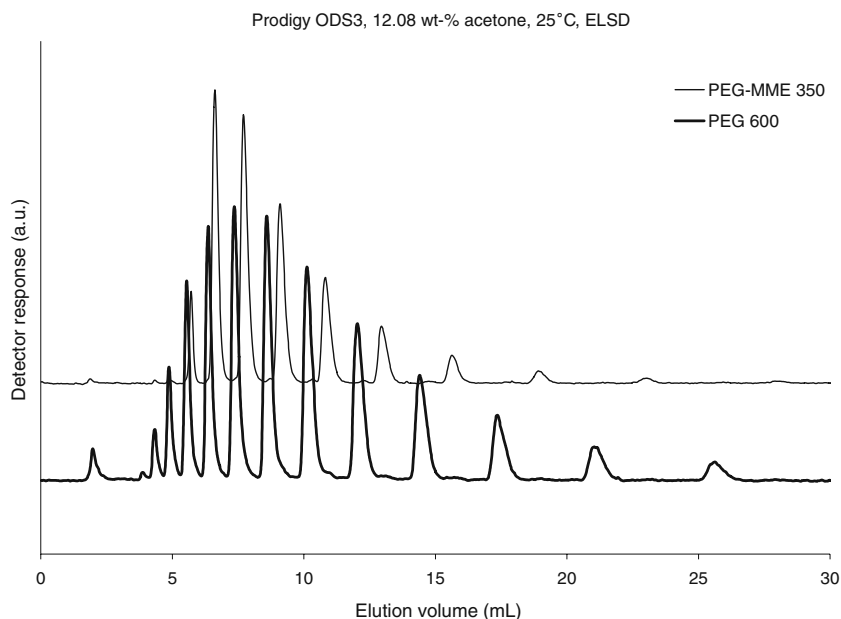
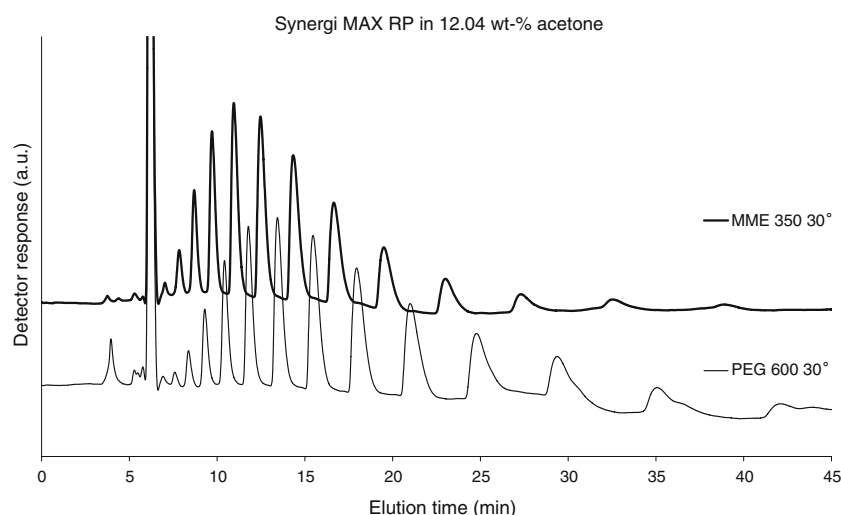


Fig. 4 LAC of PEG 600 and PEG-MME 350 on the Synergi MAX RP column in 12.04 wt.% acetone at 30 °C. Detection: RI



Obviously, pairs of oligomers of both series will coelute, if Δn is integer, and they will be separated, if $\Delta n = 0.5, 1.5, 2.5, 3.5$, etc.

This can be illustrated by a simulation. In a previous study [39], we have separated PEG and PEG-MME on a Prodigy ODS3 column in 12.04 wt.% acetone–water and obtained the result given in Table 1.

From these parameters, we have calculated the retention factors for different values of q . For the sake of simplicity, we have assumed a constant value of the interaction parameter c of the repeat unit and varied only the endgroup parameter q .

As can be seen in Fig. 1, there are pairs of peaks in both series, which coincide completely, if the value of q is unfavorable, while the series are perfectly separated, if the value of q is favorable (Fig. 2). Obviously, q and c depend on temperature and mobile phase composition. The question now is whether variation of these parameters can yield a combination corresponding to the situation shown in Fig. 2.

As a typical example, we have chosen the system PEG and PEG-MME, which has been investigated extensively in previous papers [38–41].

We have thus studied the influence of mobile phase composition and temperature on c of PEG and q of the methoxy group on two reversed-phase columns in different mobile phases (acetone–water and methanol–water).

Experimental

These investigations were performed using different systems.

System A is an isocratic HPLC system Ultimate 3000, consisting of a pump ISO-3100A, column thermostat TCC-3000, autosampler WPS-3000SL, and an RI detector RI 101, all from Dionex (Germerink, Germany). Data acquisition and processing was performed using the software Chromeleon (Dionex). The flow rate was 0.5 mL min^{-1} in all measurements.

Fig. 5 LAC of PEG 600 and PEG-MME 350 on the Synergi MAX RP column in 31.8 wt.% methanol at 30 °C. Detection: RI

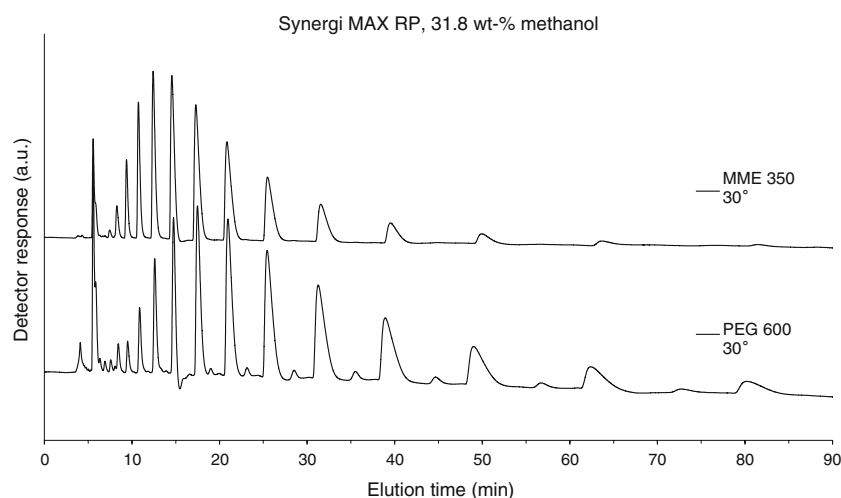
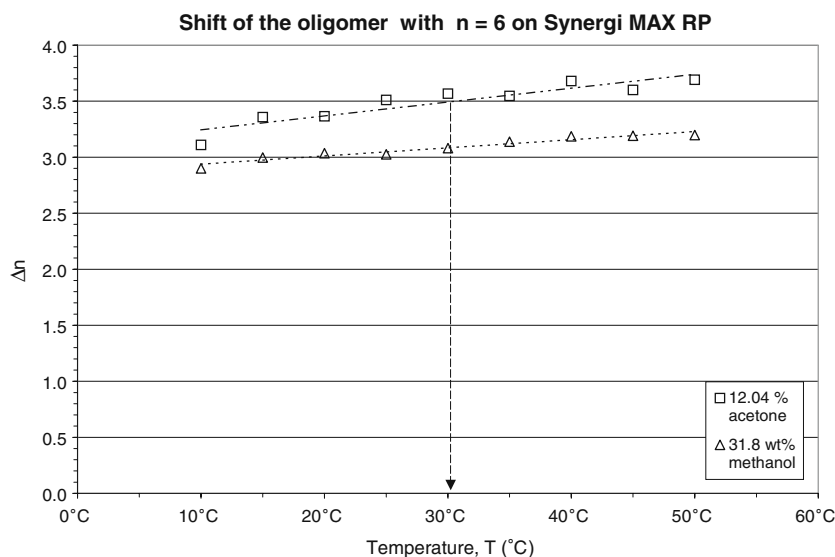


Fig. 6 Oligomer shift (Eq. 12) on Synergi MAX RP in different mobile phases



System B is a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, and autosampler WPS-3000SL, all from Dionex, an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories/Varian, Church Stretton, Shropshire, UK). All separations were performed at a flow rate of 0.5 mL min^{-1} . The mobile phases were mixed by the gradient pump, so their compositions are given in volume percent. Data acquisition and processing was performed using the software Chromeleon (Dionex).

System C was an isocratic system using the density detection system DDS70 (CHROMTECH, Graz, Austria). The mobile phase was delivered by a JASCO 880 PU pump (Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 0.5 mL min^{-1} . Samples were injected manually using a

Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a $50\text{-}\mu\text{L}$ loop. A Waters 2414 refractive index detector (Waters, Milford, MA, USA) or a SEDEX 45 ELSD (Sedere, Alfortville, France) was connected to the DDS 70.

Data acquisition and processing was performed using the software package CHROMA, which has been developed for the DDS70.

The following columns were used in this study (specifications given by the producer):

Synergi 4μ MAX RP 80A (Phenomenex, Torrance, CA, USA): silica-based dodecyl phase; $250 \times 4.6 \text{ mm}$; particle diameter = $4 \mu\text{m}$; pore size = $80 \text{ \AA}$; surface area, $475 \text{ m}^2/\text{g}$; carbon load, 17%.

Fig. 7 LAC of PEG 600 and PEG-MME 350 on the Synergi Fusion RP column in 45 vol.% methanol at 50°C . Detection: ELSD

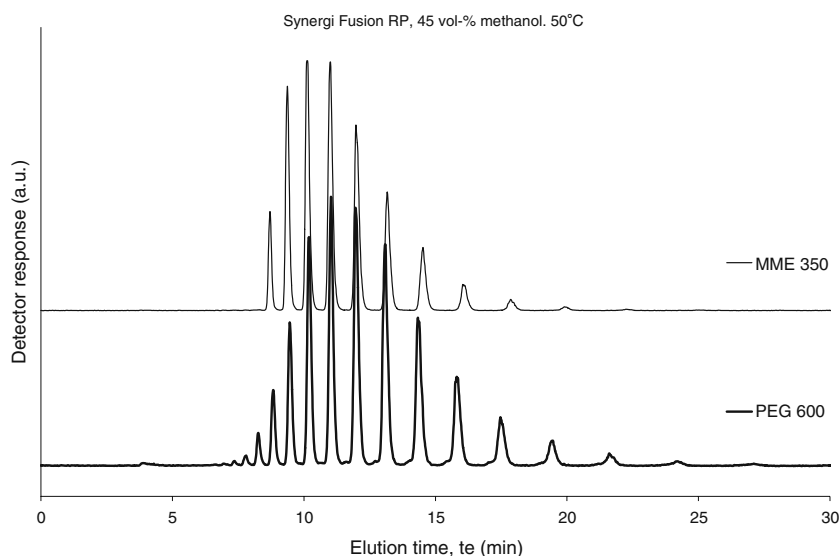
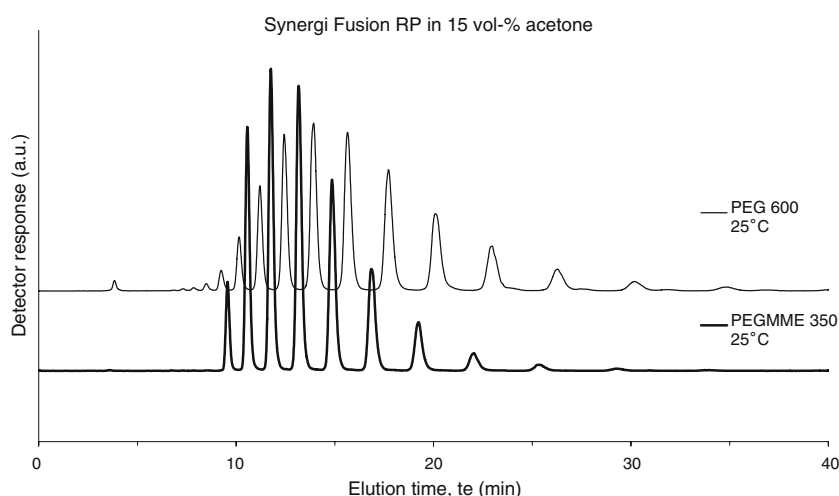


Fig. 8 LAC of PEG 600 and PEG-MME 350 on the Synergi Fusion RP column in 15 vol.% acetone at 25 °C. Detection: ELSD



Synergi Fusion RP (silica-based dodecyl phase; 250×4.6 mm; particle diameter=4 μm; nominal pore size=80 Å (Phenomenex)).

Prodigy ODS3 (Phenomenex): silica-based octadecyl phase; 250×4.6 mm; particle diameter=5 μm; pore size=100 Å, surface area, 450 m²/g; carbon load, 15.5%.

The solvents (methanol, acetone, and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany). In the gradient system, mobile phases were mixed by the gradient pump and degassed on-line. In the isocratic systems (A and C), mobile phases were mixed by mass and vacuum-degassed, and their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

PEG and their monomethyl ethers (Me-PEG) were purchased from FLUKA (Buchs, Switzerland). These samples were quite clean, as could also be confirmed by MALDI-TOF-MS.

Results and discussion

First of all, we wanted to reproduce the LCCC separation described by Barman [22] with our equipment. Critical conditions for EO were found on the Synergi Fusion RP column in 82% methanol at a temperature of 25 °C. Under these conditions, the retention times of PEGs and PEG-MMEs are sufficiently different to allow such a separation (not shown). An even better resolution can be achieved in acetone–water, as has been described in a previous paper [24].

After all, a separation of PEG and PEG-MME by LCCC can be achieved, but there remains the question of the MMD of PEG. As we have shown previously [24], this can be solved by 2D-LC, but this approach requires rather complex equipment, which is not available in many labs.

As has been discussed above, a full separation of all oligomers of both series should be possible at a special combination of the interaction parameter of the end group and the repeat unit. The dependence of c and q on mobile

Fig. 9 Martin's plot for PEG and PEG-MME on the Synergi Fusion RP column in 15 vol.% acetone at 10 °C

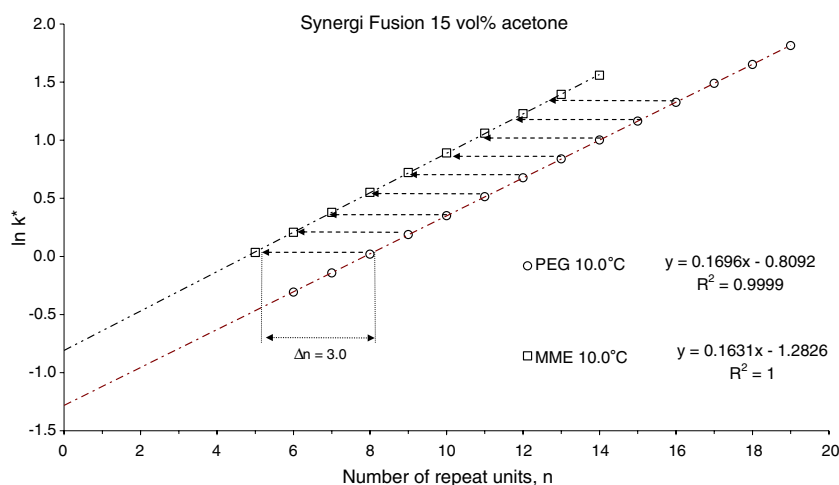
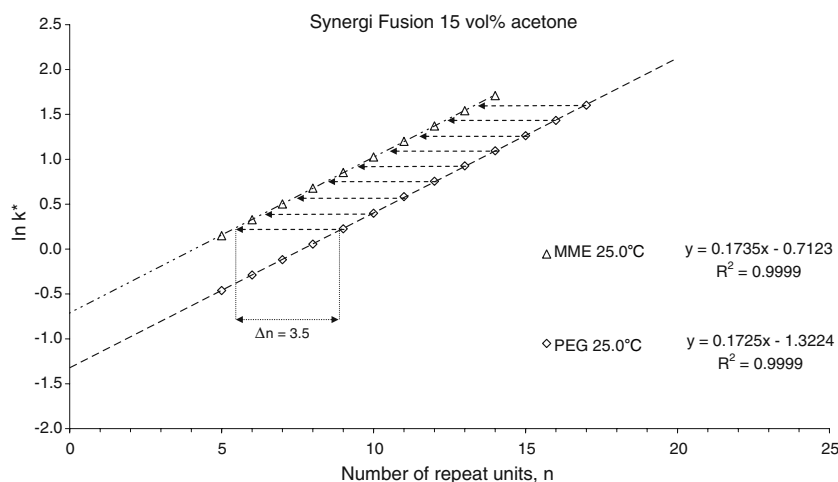


Fig. 10 Martin's plot for PEG and PEG-MME on the Synergi Fusion RP column in 15 vol.% acetone at 25 °C



phase composition and temperature has already been studied in previous papers [38, 39].

One may try to change the shift between the individual series of peaks by adjusting the parameter qc , which can be done by changing the temperature and mobile phase composition, but this will change both the interaction parameter of the repeat unit c and the end group parameter q . Obviously, a reasonable separation will only be possible in a rather narrow range of c , which means that also the available range of q is limited.

The only chance to find suitable conditions will exist in systems, in which c and q show a different dependence on mobile phase composition and temperature. In this study, we have now tried to find such systems by using the data obtained on various columns in different mobile phases. The approach described above allowed a prediction; these systems may be useful for this purpose.

Based on the data obtained with the Prodigy ODS3 column (see Table 1), we have selected conditions, under which favorable (and for comparison also unfavorable) combinations of c and q could be expected (corresponding to Figs. 1 and 2). As has been discussed previously [39, 41, 42], the interaction parameters of a given repeat unit in a given mobile phase on different columns with the same chemical nature are typically almost the same. Hence, we used three different columns in order to demonstrate the applicability of this concept.

The favorable situation is shown in Figs. 3 and 4: On the Prodigy column, the desired separation could be achieved in acetone–water (with 12.08 wt.% acetone) at 25 °C. Very similar conditions work also on the Synergi MAX RP column: In 12.04 wt.% acetone, the peaks of the PEG fall exactly between those of the PEG-MME.

The unfavorable case is shown in Fig. 5: On the Synergi MAX RP column, both series coincide perfectly in a mobile

Fig. 11 Oligomer shift (Eq. 12) on Synergi Fusion RP in acetone–water with 15 vol.% acetone

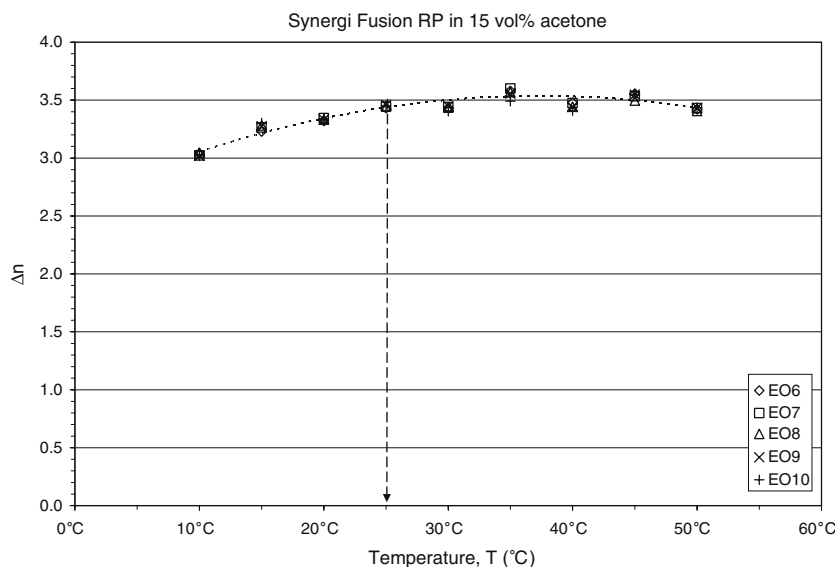
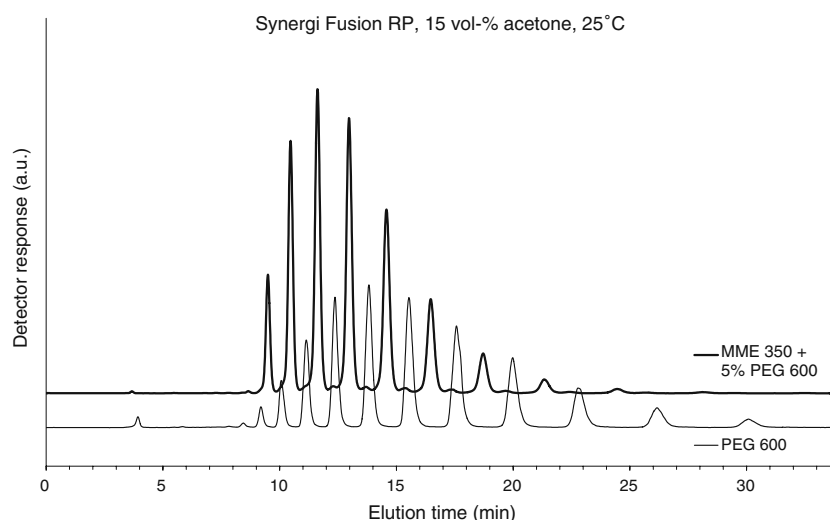


Fig. 12 LAC of PEG-MME 350 and PEG-MME 350 containing 5% PEG 600 on the Synergi Fusion RP column in 15 vol.% acetone at 25 °C. Detection: ELSD



phase containing 31.8 wt.% methanol, and no discrimination of the individual series is possible.

A variation of temperature does not help in methanol–water, but in acetone–water, it allows a fine adjustment, as can be seen in Fig. 6.

The same behavior is observed on the Synergi Fusion RP column. In the following figures, the composition of the mobile phase is given in volume percent, as these measurements were performed on the gradient system, where the mobile phases were mixed by the gradient pump.

Again, the series coincided in methanol–water (Fig. 7), and they were well resolved in acetone–water (Fig. 8). In all separations, the individual peaks were identified by using monodisperse oligomers as external or internal standards.

Temperature variation showed that the PEG oligomers elute between the PEG-MME oligomers in a range between 25 and 50 °C, while at a lower temperature, no reasonable separation of the individual series can be achieved (Figs. 9, 10, and 11).

For the analysis of samples containing small amounts of PEG, it is important that even very small peaks of PEG can be reasonably integrated. We have thus prepared solutions of PEG-MME 350 containing 2, 5, and 10 wt.% PEG 600 by mixing stock solutions of the pure samples. The purity of the commercially available samples was checked by MALDI-TOF-MS, which showed that the samples used in this studies contained only traces of other homologous series.

Fig. 13 LAC of PEG-MME 350 containing different amounts of PEG 600 and the pure samples on the Synergi MAX RP column in 12.15 wt.% acetone at 25 °C. Detection: RI

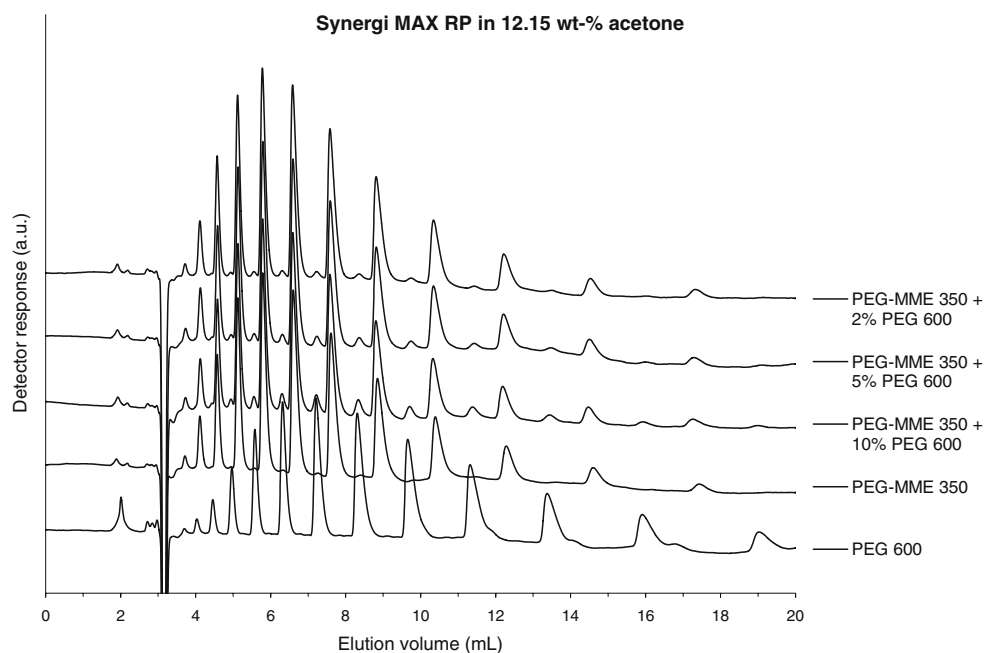


Fig. 14 Separation of PEG-MME containing 5% PEG 600 with different detectors (for conditions, see Figs. 12 and 13)

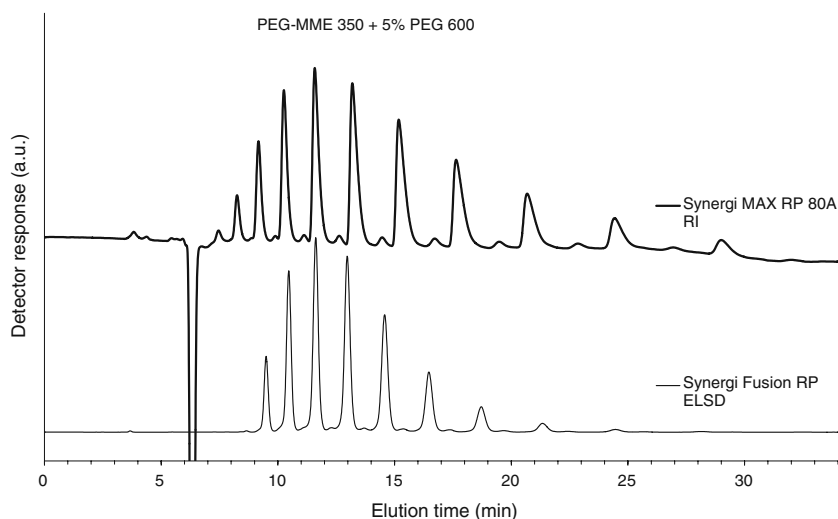


Figure 12 shows two chromatograms of pure PEG-MME 350 and a mixture with 5% PEG 600, which were obtained on the Synergi Fusion RP column in 15 vol.% acetone at 25 °C.

Evidently, a reasonable separation is achieved, but the peaks PEG oligomers appear too small. The reason is the poor linearity of the ELSD. This problem cannot be solved by increasing the sample size: Overloading the column causes much broader peaks of the PEG-MME, which start to merge with the PEG series. Moreover, the lower oligomers of PEG-MME are not detected by the ELSD even at low temperatures of nebulizer and evaporator.

Consequently, the RI detector proved to be superior to the ELSD: Even though the long heat exchanger capillary in front of the detector cell of the RI detector causes somewhat broader peaks than the ELSD, reasonable results

can be obtained even at low contents of PEG, as can be seen in the following figures.

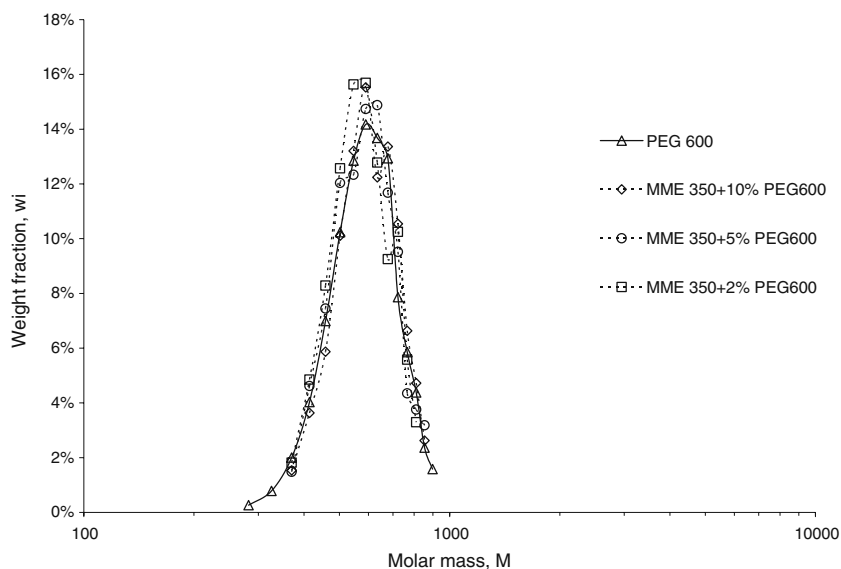
Figure 13 shows the chromatograms of mixtures of PEG-MME 350 containing 2, 5, and 10 wt.% PEG 600 and the pure samples.

As can be seen in Fig. 14, the relative sizes of the peaks correspond quite well to the mixing ratio (95:5) in RI detection, while the PEG is strongly underestimated with ELS detection because of the nonlinear response of the ELSD.

In Fig. 15, the molar mass distribution of the PEG calculated therefrom are shown. Evidently, the MMDs agree quite well.

The problem of an accurate quantification in LAC of low molecular polyethers with different functionality using different detectors is the subject of another paper [43], in

Fig. 15 MMD of the PEG fraction and the pure sample PEG 600, as obtained from the chromatograms shown in Fig. 12



which we discuss the molar mass dependence of response factors within polymer homologous series and the nonlinear behavior of the ELSD. The same problems arise as well in the 2D separation [24], where the ELSD is the only choice.

Conclusion

Different polymer homologous series with the same repeat unit, but different end groups, can be separated by LAC according to the number of repeat units and functionality by one-dimensional LAC, if a favorable combination of the interaction parameters of the repeat unit and the end groups can be realized. This can be achieved in principle by variation of mobile phase composition and temperature. If the dependencies of the interaction parameters of repeat unit and end groups are known on temperature and mobile phase composition is known, one may predict whether such a combination exists. The molar mass distribution of the minor component in such a mixture can be determined even at concentrations of a few percent. If a favorable combination of interaction parameters does not exist, two-dimensional separations with LCCC as the first and LAC as the second dimension have to be applied. In both cases, however, the crucial point in the determination of the MMD is the detector. In a separate paper [43], we shall show how accurate results for the MMD of the minor component can be obtained.

Acknowledgements H. A. thanks the Higher Education Commission of Pakistan for a Ph.D. scholarship. We also thank Prof. Robert Saf (Central Polymer Laboratories, Stremayrgasse 16, Graz, Austria) for the analysis of our PEG and PEG-MME samples by MALDI-TOF-MS.

References

- Kricheldorf HR, Ahrendorf K, Rost S (2004) *Macromol Chem Phys* 205:1602
- Simons DM, Verbanc JJ (1960) *J Polym Sci* 44:303
- Becker H, Wagner G, Stolarzewicz A (1982) *Acta Polym* 33:34
- Yu GE, Heatley F, Booth C et al (1994) *J Polym Sci Part A: Polym Chem* 32:1131
- Gallet G, Carroccio S, Rizzarelli P et al (2002) *Polymer* p 1081
- Kosswig K (1993) In: Kosswig K, Stache H (eds) Carl Hanser Verlag, München, p 115
- Lewis JL (1998) In: van Os NM (ed) *Nonionic surfactants—organic chemistry*, vol 72. Marcel Dekker, New York, p 201
- Kazanskii KS, Lapienis G, Kuznetsova VI et al (2000) *Polym Sci Series A* 42:585
- Kodera Y, Matsushima A, Hiroto M et al (1998) *Prog Polym Sci* 23:1233
- Veronese FM (2001) *Biomaterials* 22:405
- Li J, Kao WJ (2003) *Biomacromolecules* 4:1055
- Mao S, Shuai X, Unger F et al (2005) *Biomaterials* 26:6343
- Pasut G, Caboi F, Schrepfer R et al (2007) *React Funct Polym* 67:529
- Pasut G, Veronese FM (2007) *Progress in Polymer Science Polymers in Biomedical Applications* 32:933
- Yu L, Chang GT, Zhang H et al (2008) *Int J Pharm* 348:95
- Zhang J, Zhao Y-J, Zhi-Guo Su et al (2007) *J Appl Polym Sci* 105:3782
- Selisko B, Delgado C, Fisher D et al (1993) *J Chromatogr* 641:71
- Gorshkov AV, Jevreinov VV, Lausacker B et al (1986) *Acta Polym* 37:740
- Filatova NN, Gorshkov AV, Evreinov VV et al (1988) *Vysokomol Soedin, Ser A* 30:953
- Gorshkov AV, Much H, Becker H et al (1990) *J Chromatogr* 523:91
- Gorbunov A, Trathnigg B (2002) *J Chromatogr A* 955:9
- Barman BN, Champion DH, Sjöberg SL (2009) *J Chromatogr A* 1216:6816
- Trathnigg B, Kollroser M, Berek D et al (1999) Quantitation in the analysis of oligomers by HPLC with evaporative light scattering detector. *ACS Symposium Series* 731:95–112
- Ahmed H, Trathnigg B (2010) *J Sep Sci* 33:1448
- Martin AJP (1949) *Biochem Soc Symp* 3:4
- Tchaplá A, Heron S, Lesellier E et al (1993) *J Chromatogr A* 656:81
- Tchaplá A, Colin H, Guiochon G (1984) *Anal Chem* 56:621
- Krstulovic AM, Colin H, Guiochon G (1982) *Anal Chem* 54:2438
- Colin H, Krstulovic AM, Gonnord MF et al (1983) *Chromatographia* 17:9
- Skvortsov A, Trathnigg B (2003) *J Chromatogr A* 1015:31
- Trathnigg B, Skvortsov A (2006) *J Chromatogr A* 1127:117
- Trathnigg B, Jamelnik O, Skvortsov A (2006) *J Chromatogr A* 1128:39
- Skvortsov aM, Gorbunov AA (1990) *J Chromatogr* 507:487
- Gorbunov AA, Vakhrushev AV (2004) *Polymer* 45:7303
- Gorbunov AA, Skvortsov AM, Trathnigg B et al (1998) *J Chromatogr A* 798:187
- Trathnigg B, Gorbunov AA (2001) *J Chromatogr A* 910:207
- Skvortsov AM, Fleer GJ (2002) *Macromolecules* 35:8609
- Nguyen Viet C, Trathnigg B (2010) *J Sep Sci* 33:1052
- Trathnigg B, Nguyen Viet C, Ahmed H (2009) *J Sep Sci* 32:2857
- Trathnigg B, Jamelnik O (2007) *J Chromatogr A* 1146:78
- Trathnigg B, Malik MI, Nguyen Viet C et al (2008) *J Chromatogr A* 1207:122
- Trathnigg B, Malik M I, Jamelnik O, et al (2007) *Analytica Chimica Acta. Papers presented at the 3rd International Symposium on the Separation and Characterization of Natural and Synthetic Macromolecules—SCM-3* 604:39
- Trathnigg B, Ahmed H (2010) *J Separ Sci* 33:1448–1453. doi:10.1002/jssc.200900847